Social Media's Influence on News Consumption

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Table of Contents

01

Research Questions

02

Data Collection

03

Statistical Tests

04

Future Research and Limitations

Introduction

- Social media platforms have become central sources of news, especially among college students
- They shape how users access news and influence perceptions of credibility, trust, and knowledge
- The purpose of our project is to examine how social media usage patterns relate to news consumption, trust, misinformation, and knowledge of current events

Research Questions

1. Does frequency of social media use predict reliance on social media for news?
2. Is TikTok use related to higher exposure to misinformation?
3. Do students who fact-check more trust social media news less?
4. Are there gender differences in trust of social media news?
5. Does exposure to news on social media increase perceived knowledge of current events?

Qualtrics Survey

- Total respondents: 44
  - Ages 18-23
- Female respondents: 34
- Male respondents: 10
- 13 Likert-Scale Questions
- 2 multiple choice questions (nominal)

Scale variables

Variable	Mean	Standard Deviation	N
Frequency of Social Media Use	6.48	1.36	44
Reliance on Social Media for News	4.23	1.33	44
Exposure to Misinformation	5.34	1.40	44
Fact-checking Frequency	4.65	1.17	44
Trust in Social Media News	3.48	1.52	44
Perceived Knowledge of Current Events	4.18	1.39	44

1. Does frequency of social media use predict reliance on social media for news?

Linear Regression- Analysis

Model Summary

R-square =.205.

The three predictors together explain 20.5% of the variation in relying on social media for news. This is a moderate amount.

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.453[a]	.205	.142	1.557

a. Predictors: (Constant), Tiktok_use, Twitter_use, Instagram_Use

Linear Regression- Analysis

ANOVA table

p= .032

This means the overall model is statistically significant

Frequency of platform use does significantly predict reliance on social media for news.

Which platforms significantly predict reliance…

Instagram - .564 Not Significant

Twitter - .042 Significant

TikTok - .031 Significant

People who use Twitter and TikTok more frequently are more likely to rely on social media as their primary news source.

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	23.774	3	7.925	3.269	.032[b]
	Residual	92.130	38	2.424		
	Total	115.905	41			

a. Dependent Variable: SocialMedia_PrimarySource

b. Predictors: (Constant), Tiktok_use, Twitter_use, Instagram_Use

Coefficients[a]

Model		Unstandardized Coefficients B	Unstandardized Coefficients Std. Error	Standardized Coefficients Beta	t	Sig.
1	(Constant)	1.292	1.480		.873	.388
	Instagram_Use	.108	.186	.088	.582	.564
	Twitter_use	.382	.182	.314	2.104	.042
	Tiktok_use	.327	.146	.327	2.240	.031

a. Dependent Variable: SocialMedia_PrimarySource

Linear Regression- Analysis

Instagram

Mean= 6.18

SD = 1.386

Twitter

Mean= 1.60

SD = 1.380

TikTok

Mean= 6.23

SD = 1.655

Descriptive Statistics

	N	Minimum	Maximum	Mean	Std. Deviation
Instagram_Use	44	1	7	6.18	1.386
Twitter_use	42	1	7	1.60	1.380
Tiktok_use	44	1	7	6.23	1.655
SocialMedia_PrimarySource	44	1	7	4.59	1.703
Valid N (listwise)	42				

How much they rely on social media for news

Mean reliance on social media for news = 4.56

On average, people somewhat rely on social media for news, rating it slightly above the midpoint of the scale.

Marketing Implication

Focus on TikTok and Twitter

- These platforms are better at changing what people pay attention to.
- Higher Twitter (.042 Sig.) and TikTok (.032 Sig.) use significantly increases reliance on social media as a primary news source
- People are more likely to make TikTok or Twitter as their main social media.

Use Instagram as support

- Instagram use is high, but it does not influence primary platform choice (.564 Sig.)
- Do not use as the primary source of news for persuasion or loyalty

Spend money wisely

- Invest more in TikTok and Twitter ads or strategies since they show stronger behavioral impact.
- Just because Instagram is widely used does not mean it is the most influential.

To influence where consumers focus their social media attention, marketers should emphasize TikTok and Twitter, using Instagram mainly as a complementary branding platform rather than a primary driver.

2. Is TikTok use related to higher exposure to misinformation?

Pearson Correlation- Analysis

Descriptive Statistics

- **Tik Tok mean= 6.23**. Very high on a 1-7 scale. Most students in our sample use Tik Tok many times a day.
- **Standard Deviation= 1.655**, there is some variation in use but overall students lean heavily toward frequent use.

- **Misinformation concern mean= 5.34**. This is also high. Students generally worry about misinformation on TikTok and other platforms.
- **Standard Deviation= 1.397**, most students fall close to the same level of concern.

Descriptives

Descriptive Statistics

	N	Minimum	Maximum	Mean	Std. Deviation
Tiktok_use	44	1	7	6.23	1.655
Misinformation_Concern	44	1	7	5.34	1.397
Valid N (listwise)	44				

Pearson Correlation- Analysis

Correlations

- Correlation between Tik Tok use and Misinformation Concern is **0.237.**
  - Weak positive relationship.
  - Students who use Tik Tok more show only a small increase in how much they worry about misinformation.

- Significant value point is **0.121>.05** the result is not significant.

Correlations

		Tiktok_use	Misinformation_Concern
Tiktok_use	Pearson Correlation	1	.237
	Sig. (2-tailed)		.121
	N	44	44
Misinformation_Concern	Pearson Correlation	.237	1
	Sig. (2-tailed)	.121	
	N	44	44

Marketing Implication

- Weak correlation of **0.237** means heavy TikTok users are not much more worried about misinformation. Marketers can still target them normally.

- The high average TikTok use of **6.23** shows students are very active on the app so Tik Tok remains a strong place for marketing.

- The concern average of **5.34** shows credibility still matters. Clear sources and transparent messaging will matter across all user groups.

3. Do students who fact-check more trust social media news less?

Descriptives

- The descriptive statistics show that fact-checking behavior had a mean of **4.45** (SD = 1.73) and trust in social media news had a mean of **3.48** (SD = 1.52), both measured on a 1–7 scale.

Descriptives

Descriptive Statistics

	N	Minimum	Maximum	Mean	Std. Deviation
Fact_Checking	44	1	7	4.45	1.731
Trust_SocialMedia_News	44	1	7	3.48	1.517
Valid N (listwise)	44				

Pearson Correlation

- Pearson correlation: -.244
- P-value (Sig. 2-tailed): .111

The correlation between fact checking and trust in social media news is –0.24

- This means there is a weak negative relationship.
- As fact checking goes up, trust in social media news tends to go slightly down.

The p-value is .111, which is not statistically significant.

So we can not conclude that higher fact-checking actually predicts lower trust in social media news

Correlations

		Fact_Checking	Trust_SocialMedia_News
Fact_Checking	Pearson Correlation	1	-.244
	Sig. (2-tailed)		.111
	N	44	44
Trust_SocialMedia_News	Pearson Correlation	-.244	1
	Sig. (2-tailed)	.111	
	N	44	44

Marketing Implication

- Weak trust in social media news suggests marketers need to prioritize credibility
- Students show low trust in social media news (mean= 3.48) so marketers should focus on building authenticity and transparency
- Fact-checkers are still reachable consumers, considering the weak correlation, they just need more evidence and credibility
- Brands can benefit from offering clear data or showing behind-the-scenes processes to gain trust
- If consumers do distrust social media, companies can consider using platforms other than social media, like email, websites, or traditional media

4. Are there gender differences in trust of social media news?

Descriptive Statistics-Analysis

1. Group Means

- **Males (n = 10):** Mean = **3.20**, SD = 1.75
- **Females (n = 34):** Mean = **3.56**, SD = 1.46

Descriptive Statistics[a]

Gender		N	Minimum	Maximum	Mean	Std. Deviation
male	Trust_SocialMedia_News	10	1	7	3.20	1.751
	Gender	10	1	1	1.00	.000
	Valid N (listwise)	10				
female	Trust_SocialMedia_News	34	1	7	3.56	1.460
	Gender	34	2	2	2.00	.000
	Valid N (listwise)	34				

Interpretation:

Females reported *slightly higher trust* in social media news compared to males, but the difference is small.

Group Statistics

	Gender	N	Mean	Std. Deviation	Std. Error Mean
Trust_SocialMedia_News	male	10	3.20	1.751	.554
	female	34	3.56	1.460	.250

Independent Samples t-Test

Levene's Test

- Sig. = .787
- Because .787 > .05, use the "Equal variances assumed" row

p (two-sided) = .517

- The p-value (.517) is much higher than .05, meaning the difference between genders is not statistically significant.

Independent Samples Test

		Levene's Test for Equality of Variances		t-test for Equality of Means							
						Significance				95% Confidence Interval of the Difference	
		F	Sig.	t	df	One-Sided p	Two-Sided p	Mean Difference	Std. Error Difference	Lower	Upper
Trust_SocialMedia_News	Equal variances assumed	.074	.787	-.653	42	.259	.517	-.359	.549	-1.468	.750
	Equal variances not assumed			-.590	12.911	.283	.565	-.359	.608	-1.673	.955

Independent Samples Effect Sizes

- Cohen's d = **–.235** (very small effect)
- **Interpretation:**
  The effect of gender on trust is small and weak, meaning gender does not meaningfully explain differences in trust.

Independent Samples Effect Sizes

		Standardizer[a]	Point Estimate	95% Confidence Interval Lower	Upper
Trust_SocialMedia_News	Cohen's d	1.527	-.235	-.940	.473
	Hedges' correction	1.555	-.231	-.923	.465
	Glass's delta	1.460	-.246	-.951	.464

a. The denominator used in estimating the effect sizes.
Cohen's d uses the pooled standard deviation.
Hedges' correction uses the pooled standard deviation, plus a correction factor.
Glass's delta uses the sample standard deviation of the control (i.e., the second) group.

Marketing Implication

- Gender-targeted messaging isn’t necessary most of the time, they show similar trust levels
- Focus segmentation on behavior, not gender because gender doesn’t predict trust
- Since gender isn’t significant, marketers should focus on:
  - Improving message clarity
  - Using trusted spokespeople
  - Reducing misinformation risks
  - Highlighting authenticity

5. Does exposure to news on social media increase perceived knowledge of current events?

Pearson Correlation- Analysis

Descriptive Statistics

- **News exposure mean= 5.05** students see news on social media fairly often.
- **Standard deviation= 1.555** their exposure levels vary quite a bit from person to person.

- **Perception influence mean=4.80** students feel social media influences their views at a moderate level.
- **Standard deviation= 1.593** shows influence levels also vary across the group.

Descriptive Statistics

	N	Minimum	Maximum	Mean	Std. Deviation
News_Exposure_NewsSource	44	1	7	5.05	1.555
Perception_Influence	44	1	7	4.80	1.593
Valid N (listwise)	44				

Pearson Correlation - Analysis

How strong the relationship is between news exposure and perceived influence.

R = .314

- This is a moderate positive relationship

R Squared = .098

- News exposure explains about 10% of how influenced people feel

Model Summary[b]

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.314[a]	.098	.077	1.531

a. Predictors: (Constant), News_Exposure_NewsSource

b. Dependent Variable: Perception_Influence

Pearson Correlation – Analysis

Sig. (p-value) = .038

- Because this is below .05, the result is statistically significant.

As people are exposed to more news, they feel more influenced by it.

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	10.742	1	10.742	4.584	.038[b]
	Residual	98.417	42	2.343		
	Total	109.159	43			

a. Dependent Variable: Perception_Influence

b. Predictors: (Constant), News_Exposure_NewsSource

Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.	95.0% Confidence Interval for B	
		B	Std. Error	Beta			Lower Bound	Upper Bound
1	(Constant)	3.173	.792		4.006	<.001	1.575	4.772
	News_Exposure_NewsSource	.322	.150	.314	2.141	.038	.018	.625

Marketing Implication

News alone will not change behavior

Since most influence comes from other factors (news exposure explains only about 10% of perceived influence)

- News should not be the only channel
- Social media, peers, and personal beliefs also matter

Consistent exposure matters

- Higher news exposure is significantly associated with feeling more influenced ($p = .038$), marketers should recognize that frequent exposure matters.
- Repeated visibility of messages can gradually shape how audiences think and feel

Limitations

Limitations

- **Self Reported Survey Data Collection**
  - People might not remember their behavior clearly or may answer in a way they think makes them look better
- **Small Sample Size**
  - Total of 44 College Students (34 Female, 10 Male)
  - Some relationships may exist but did not reach significance simply because there were not enough respondents. College students also consume news differently from older adults.
- **Focused On Three Platform**
  - We focused on Instagram, Twitter, Tik Tok. Our study does not capture all of social media platforms

Limitations

- **Question Interpretation**
  - Participants may interpret survey questions differently, leading to measurement error.
  - What counts as “news exposure” or how often is “often”
- **Scale Constraints**
  - Using Likert-type scales (1–7) may oversimplify complex attitudes and behaviors
  - limiting participants’ responses.

Future Research

Future Research

- Use a larger and more representative sample to include a wider age range and diverse demographic groups. This would allow for a more accurate picture of social media news consumptions across different demographics.
- Tracking participants over time to see how changes in social media habits influence trust, misinformation exposure, and perceived knowledge
- Incorporate screen time or engagement logs on social media instead of relying only on self-reported survey data
- Compare results with actual knowledge tests to see if social media increases actual knowledge or just perceived knowledge
- Include more demographic questions such as academic major, level of education, etc.

Thank you!

